

Agentic Coding with CUDA-Q

From quantum basics to verified, GPU-accelerated applications

THE WORKING RHYTHM **PROMPT** › **BUILD** › **RUN** › **VERIFY** One focused step at a time, with a working result before moving on.

YOUR JOURNEY

- 0 Meet CUDA-Q** NO AGENT YET
 Explore states, gates, measurement, entanglement, kernels, and variational algorithms. _intro_cudaq.ipynb
- 1 Equip the coding agent** CODING AGENT
 Give the agent a CUDA-Q reference plus the course rules for reproducible work. cudaq-doc.md + AGENTS.md
- 2 Create a GHZ quick win** CODING AGENT
 Turn a plain-language prompt into a 20-qubit GPU program and verify its counts. create_ghz.py
- 3 Meet the calibration agent** CALIBRATION AGENT
 Simulate an experiment, attach its plot, and ask an expert model to interpret it. _intro_Ising_Calibration.ipynb
- Q Advance a QAOA application** CODING AGENT
 Optimize and scale a verified Taiwan map-coloring workflow from 16 to 36 qubits. 00_notebook.ipynb › 04_notebook.ipynb

NVIDIA TECHNOLOGY IN ACTION

CORE PLATFORM CUDA-Q Build, simulate, optimize, sample, and verify quantum programs.	PHYSICS SIMULATION CUDA-Q Dynamics Simulate open-system calibration experiments on an NVIDIA GPU.	VISION-LANGUAGE MODEL NVIDIA Ising Calibration Read calibration plots and return structured technical analysis.	CODING-AGENT MODEL Nemotron 3 Ultra 550B A55B Power the optional OpenCode coding-agent workflow through NVIDIA NIM.
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BEFORE YOU BEGIN

- Python 3.12
- CUDA-Q 0.15.0
- 1 NVIDIA GPU
- 1 coding agent

Supporting packages install from requirements.txt; the bootcamp image provides CUDA-Q.

BY THE END

You will have a repeatable way to co-work with agents

You will know how to choose a simulator, improve GPU execution, verify sampled answers, analyze calibration plots, and carry the method into your own CUDA-Q project.

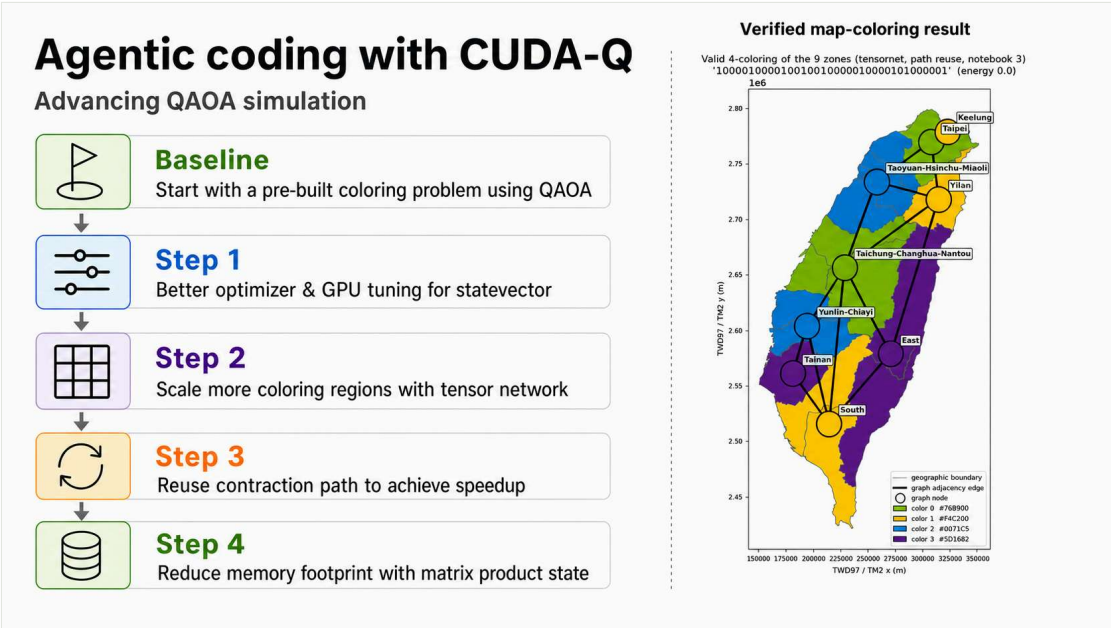
Advance QAOA, step by step

One map-coloring problem becomes a practical GPU optimization ladder

STAGE	FILES USED / CREATED	WHAT YOU LEARN	NVIDIA TECH + RESULT
BASE	Use: 00_notebook.ipynb helpers.py + map data	Turn map rules into a QUBO; build QAOA; compare CPU and GPU.	qpp-cpu vs CUDA-Q nvidia FP32 Verified 16-qubit coloring
STEP 1	Read: 00 + AGENTS.md Create: 01_notebook.ipynb	Swap to Nelder-Mead, tune GPU execution, and time each iteration.	CUDA-Q nvidia FP32 Better energy and valid rate
STEP 2	Read: 00 / 01 + helpers Create: 02_notebook.ipynb	Scale 16 to 36 qubits with tensor networks and an XY mixer.	CUDA-Q tensornet FP32 9 zones beyond state-vector memory
STEP 3	Read: 02_notebook.ipynb Create: 03_notebook.ipynb	Reuse the contraction path instead of planning it every time.	CUDA-Q tensornet FP32 Same verified answer, less overhead
STEP 4	Read: 03_notebook.ipynb Create: 04_notebook.ipynb	Use matrix product state compression; understand the trade-off.	CUDA-Q tensornet-mps FP32 Lower memory; samples verified exactly

CORE MISSION

Color a Taiwan map so neighboring areas differ — then scale from a 16-qubit baseline to verified 36-qubit simulations.



The ladder at a glance, with the verified four-coloring of nine Taiwan zones. COURSE VISUAL

Files, agent roles, and your take-home toolkit

Know what to open, what to build, and what you will carry forward

TWO AGENTS, TWO JOBS

CODE CREATION

Coding agent

OpenCode, Claude, or Codex reads your files, writes CUDA-Q code, runs it, and reports verified results.

Optional OpenCode model: NVIDIA Nemotron 3 Ultra 550B A55B via NVIDIA NIM

CALIBRATION ANALYSIS

Calibration agent

NVIDIA Ising Calibration reads experiment plots, checks fit quality, extracts parameters, and suggests corrective action.

Notebook: `_intro_Ising_Calibration.ipynb` · Simulation: CUDA-Q Dynamics

FILE MAP

requirements.txt

Supporting Python packages.

_intro_cudaq.ipynb

States, gates, measurement, entanglement, kernels, and VQAs.

cudaq-doc.md TAKEAWAY

The agent-ready CUDA-Q reference.

AGENTS.md

Pacing, reproducibility, validation, and notebook rules.

_intro_Ising_Calibration.ipynb

Dynamics simulation plus calibration-plot analysis.

helpers.py + data/taiwan_map_xy.json

Map problem, decoding, verification, and visualization.

00_notebook.ipynb

The supplied QAOA baseline.

01_notebook.ipynb ... 04_notebook.ipynb

The participant-built GPU optimization ladder.

my_code.py

A Qiskit Grover example for BYOC migration.

SKILL.md TAKEAWAY

Your reusable CUDA-Q GPU optimization method.

cudaq-guide TAKEAWAY

Official NVIDIA CUDA-Q onboarding skill.

WHAT YOU TAKE HOME

- A verified GPU quantum-programming workflow.
- A playbook for optimizer, backend, memory, and speed choices.
- Experience separating code creation from calibration analysis.
- A path to migrate existing code or build a new CUDA-Q idea.
- A reusable SKILL.md for future CUDA-Q projects.



SCAN FOR BOOTCAMP MATERIALS